Computer Security — Practice Midterm Test

MAIN TEST · Multiple choice · Weeks 1–6 · 60 questions × 1 mark = 60 marks · Suggested time: 60 minutes

Name: _____	**Student ID:** _____	**Score:** _____ / 60

> **Instructions.** Choose the **one best** answer for each question and circle the letter. The questions follow the lecture order (week 1 → week 6). The answer key with a one-line reason for every question is on the last pages. Try the whole test before you look.

Week 1 — Introduction to Computer Security

1. Which statement best matches the course definition of **computer security**?

A. Software that removes viruses from a computer
B. Ideas, practices and controls that reduce unauthorized access, unwanted change and harmful disruption
C. Encrypting every file stored on a device
D. Rules that stop people from using the Internet

2. A student's grades are accidentally posted on a public notice board. Which security goal failed?

A. Integrity
B. Availability
C. Confidentiality
D. Authorization

3. A staff member edits the wrong row in a spreadsheet and replaces correct grades with wrong ones. No attacker was involved. Which statement is correct?

A. It is an integrity failure, even though it was an accident
B. It is not a security problem because there was no attacker
C. It is only a confidentiality failure
D. It is only an availability failure

4. In Mina's university-account case, a **reused password** is an example of which of the five words?

A. Asset
B. Threat
C. Control
D. Vulnerability

5. Which activity is **authorized** (ethical) practice?

A. Scanning a university server because it is reachable from the dorm network
B. Scanning an instructor-provided lab machine during the scheduled exercise, using approved steps, and reporting findings privately
C. Logging into a friend's account with their old password to prove it is weak
D. Posting a discovered weakness on social media to warn other students

6. What does **defense in depth** mean?

A. Buying the strongest firewall available and relying on it
B. Hiding the network so attackers cannot find it
C. Using several layers of protection, because any single control can fail
D. Training users instead of using technology

7. What is the correct safe response to a suspicious message?

A. Click, check, report
B. Report, pause, check
C. Check, reply, report
D. Pause, check, report

Week 2 — Assets, Threats and Risk

8. Two identical computers cost the same. One shows public notices; the other supports urgent hospital care. What does this show?

A. Value comes from what the asset does, not from what it costs
B. Both computers have exactly the same value
C. The cheaper-looking computer is always less valuable
D. Computers are not assets; only information is

9. In the Facebook outage (4 October 2021) the computers were fine, but a network setting change made services fail. Which asset question does this illustrate?

A. Value
B. Dependency
C. Confidentiality
D. Exploit

10. In the Equifax 2017 case, the **unfixed software flaw** is the _____ and **using the flaw to get in** is the _____.

A. threat; control
B. exploit; vulnerability
C. vulnerability; exploit
D. control; threat

11. In GitLab 2017, database data was deleted by mistake. What kind of threat is this?

A. Malicious
B. Physical
C. Social engineering
D. Accidental

12. Which case is the course's example of a **physical** threat?

A. Kakao data-center fire, 2022
B. WannaCry, 2017
C. GitLab database deletion, 2017
D. Uber MFA fatigue, 2022

13. What is the main lesson of GitLab 2017 about backups?

A. Backups are unnecessary if staff are careful
B. Recovery must be tested; a backup you never restored from is an assumption
C. Backups should stay connected to the main system at all times
D. Backups prevent every integrity failure

14. Which statement describes a **risk**, not an incident?

A. Ransomware encrypted the lab computers last night
B. A student account was used to send spam yesterday
C. The exam system has an unfixed flaw that could be used before next month's exam
D. Staff entered their passwords on a fake site this morning

15. Likelihood = 2 (Possible) and Impact = 3 (Severe). What are the score and level?

A. 5, Medium
B. 6, High
C. 6, Medium
D. 9, High

16. Which score can **never** appear in the 3 × 3 risk matrix?

A. 4
B. 6
C. 9
D. 5

17. A tested restore mainly reduces which part of risk?

A. Impact
B. Likelihood
C. The number of attackers
D. The value of the asset

18. Staff at Cloudflare (2022) entered their details on a fake site, but the attacker still could not log in. Why?

A. Staff never actually typed their details
B. The company used SMS codes
C. Origin-bound security keys checked which website was asking
D. A firewall blocked the phishing e-mail

19. Which sign-in combination is real **MFA**?

A. Password + security question
B. PIN + password
C. Two different passwords on two screens
D. Password + authenticator-app code

Week 3 — Social Engineering and Phishing

20. Social engineering mainly attacks:

A. The human factor: attention, emotion, habits and working situation
B. Bugs in operating-system code
C. Weak Wi-Fi encryption
D. Hardware failures

21. "Your scholarship payment failed" is the _____; "send me your password" is the _____.

A. requested action; pretext
B. pretext; requested action
C. lever; exploit
D. threat; control

22. "I am calling from the court. You have ten minutes to respond." Which two levers are used?

A. Reward and curiosity
B. Familiarity and secrecy
C. Authority and urgency
D. Fear and reward

23. A visitor asks you to hold a secure door so they can follow you into a restricted area. This is:

A. Baiting
B. Vishing
C. Smishing
D. Tailgating

24. An SMS uses your project name, claims to be your tutor and asks you to sign in through a link. Which labels fit best?

A. Smishing + spear phishing + impersonation
B. Vishing only
C. Broad phishing only
D. Baiting + tailgating

25. In the Uber MFA-fatigue case, what changed between the rejected prompts and the one that was accepted?

A. The request became more legitimate
B. The person's state: tired and under pressure
C. The attacker obtained a hardware security key
D. MFA was switched off by the company

26. What is the main lesson of the Hong Kong deepfake meeting (reported 2024)?

A. Video calls are always safe
B. Only e-mails can be faked
C. Recognition is not authorization
D. HTTPS prevents deepfakes

27. Which URL really belongs to `campus.example`?

A. `https://campus.example.verify-now.test/login`
B. `https://login-campus.example.net/login`
C. `https://campus-example.com/login`
D. `https://login.campus.example/student`

28. Why is the HTTPS padlock weak evidence that a website is legitimate?

A. A phishing site can also use HTTPS
B. HTTPS is not a form of encryption
C. Only government websites can use HTTPS
D. The padlock means the site contains malware

29. In South Korea, you already **sent money** after a fake "daughter had an accident" call. What should you do first?

A. Contact 118 (KISA) only
B. Call 112 and your bank's official contact to request payment suspension
C. E-mail the university IT office
D. Call the scammer back to ask for a refund

30. You received a suspicious SMS asking you to install an app. Which Korean route gives technical advice about the message or app?

A. 112
B. The university library
C. 118 (KISA)
D. The phone number given in the SMS

Week 4 — Identity, Authentication and Access Control

31. Mina types the username `mina.student`. Which step is this?

A. Authorization
B. Authentication
C. Accounting
D. Identification

32. A stranger signs in with Mina's stolen password. Which security check accepted the stranger?

A. Authentication
B. Authorization
C. Segmentation
D. Firewall filtering

33. What is **credential stuffing**?

A. Trying likely passwords such as `123456`
B. Reusing a username and password pair that leaked from another site
C. Phishing someone by voice call
D. Writing passwords on sticky notes

34. Why can login rate limits **fail** to stop offline guessing?

A. Rate limits only apply to administrator accounts
B. Offline guessing uses MFA codes instead of passwords
C. The guesses are tested against a copied hash on the attacker's own device
D. Password hashes can be decrypted instantly

35. In the CyberChef activity, the candidate `campus123` produced the target SHA-256 value. What happened?

A. The hash was decrypted
B. The salt revealed the password
C. SHA-256 is reversible
D. Hashing the guessed word produced the same hash: the guess matched

36. What is the purpose of a unique random **salt** for each password?

A. Identical passwords do not produce the same stored hash
B. It keeps the password secret, so the salt must also be secret
C. It makes hashing faster
D. It replaces the need for MFA

37. Which password best follows NIST SP 800-63B-4 for **password-only** sign-in?

A. `Campus2026!`
B. `P@ssw0rd1`
C. A randomly generated 20-character password stored in a password manager
D. An 8-character password with one symbol

38. Mina (a student) can read Joon's grade record. Which access failure is this?

A. Vertical
B. Horizontal
C. Authentication
D. Availability

39. Under **least privilege**, a TA assigned to grade section A should:

A. Edit grades for section A only, with no access to section B
B. Edit grades for every section in the course
C. Only read grades, never edit them
D. Have the same access as the lecturer

40. In the PortSwigger "2FA simple bypass" lab, the account page opened without a verification code. What failed?

A. The password was too short
B. The SMS provider was offline
C. The user approved a push prompt
D. The application did not check that MFA was completed before serving the protected page

Week 5 — Malware and Endpoint Security

41. Malware is identified mainly by:

A. Its file name
B. Its purpose and actions
C. Symptoms alone, such as a slow laptop
D. Its file size

42. Several unpatched computers become infected without any user opening a file. Most likely type?

A. Virus
B. Trojan
C. Worm
D. Spyware

43. A "free game" installs a hidden remote-control program. Most likely type?

A. Trojan
B. Worm
C. Ransomware
D. Bot

44. A program secretly records everything a student types. Most likely type?

A. Bot
B. Ransomware
C. Virus
D. Spyware (keylogger)

45. Why may a **connected** backup fail to protect you from ransomware?

A. Connected backups are slower
B. The ransomware can encrypt the connected backup too
C. Backups are not allowed on university computers
D. Backups only protect confidentiality

46. After an infection is cleaned, what can happen if you **only** restore the files?

A. The system is guaranteed to be in a trusted state
B. You no longer need to change any passwords
C. You may restore the original weakness if the cause was not fixed
D. All backups are deleted automatically

47. Which control best reduces the **excess privilege** infection path?

A. Using standard (non-administrator) accounts for daily work
B. Turning off the screen lock
C. Using the same password everywhere
D. Allowing auto-run for USB devices

48. What is the correct order of the infection story used in week 5?

A. Execution → Initial contact → Impact → Privilege → Persistence → Recovery
B. Impact → Execution → Recovery → Initial contact → Privilege → Persistence
C. Initial contact → Persistence → Recovery → Execution → Impact → Privilege
D. Initial contact → Execution → Privilege → Persistence → Impact → Recovery

49. WannaCry (2017) is best described as:

A. A virus that needed users to share infected files
B. Ransomware that also spread as a worm through a vulnerable SMB service
C. Spyware that only recorded keystrokes
D. A phishing e-mail campaign without malware

Week 6 — Networking Fundamentals for Security

50. Which statement about switches and routers is correct?

A. A switch connects different networks; a router forwards frames inside a LAN
B. Switches and routers do exactly the same job
C. A router only provides Wi-Fi
D. A switch forwards frames inside a LAN using MAC addresses; a router forwards IP packets between networks

51. What does HTTPS protect?

A. The content in transit between the two TLS endpoints
B. Your data from malware already on your own device
C. Your IP address and traffic timing
D. It guarantees that the website is honest

52. Guests in a small office should reach the Internet but **not** the staff records server. Which control fits best?

A. Antivirus on guests' phones
B. Hiding the Wi-Fi name
C. Network segmentation with rules: guest → Internet allow, guest → staff server deny
D. Changing the DNS server

53. In `198.51.100.20:22`, the number `22` usually identifies which service?

A. HTTPS
B. SSH
C. DNS
D. The device number on the LAN

54. What does DNS do?

A. Encrypts web pages
B. Blocks unwanted traffic using rules
C. Forwards frames between switch ports
D. Finds the IP address that matches a domain name

55. What is an **evil twin**?

A. A fake access point imitating a trusted Wi-Fi name
B. A virus with two copies of itself
C. Two routers with the same public IP
D. A backup copy of a server

56. Which statement about an IP address is correct?

A. It proves who the person using the device is
B. It is the same as a MAC address
C. It identifies a network interface, not a person
D. It is only used inside one LAN

57. What does **NAT** allow?

A. Encrypting all traffic leaving a network
B. Many private addresses to share one public address
C. A switch to learn MAC addresses
D. A DNS resolver to answer faster

58. The campus Internet connection fails. Can you still print to the lab printer connected to the same switch?

A. No, nothing works without the Internet
B. No, printing always goes through the LMS server
C. Only if you use HTTPS
D. Yes, because the print job stays on the local network

59. At a café you see a certificate warning on a site you use every day. What is the best first reaction?

A. Stop: do not continue, because the problem could be the device, the network or the website
B. Click "continue anyway", because you know the site
C. Turn off the antivirus and try again
D. Type your password quickly before the connection closes

60. Why does **DNS spoofing** not automatically defeat a correct HTTPS connection?

A. DNS is encrypted by the switch
B. Browsers never use DNS for HTTPS sites
C. The fake server still needs a valid certificate for that name
D. DNS spoofing only affects printers

Answer Key

Q	Ans	Why
1	B	Security reduces unauthorized access (C), unwanted change (I) and harmful disruption (A).
2	C	Private data was seen by people who should not see it.
3	A	Integrity can fail by accident; the data is no longer trustworthy.
4	D	A reused password is a weakness that could be used.
5	B	Clear owner, scope, time, method and purpose. "Reachable" ≠ "allowed".
6	C	Several layers, because any single control can fail.
7	D	Pause → check (via a route you choose) → report (official channel).
8	A	Same price, different value: value comes from what the asset does.
9	B	Services that share one dependency fail together.
10	C	Vulnerability = the flaw; exploit = its use. The attackers are the threat.
11	D	Unintended harm = accidental threat.
12	A	M / A / P: WannaCry = malicious, GitLab = accidental, Kakao fire = physical.
13	B	A backup is only proven by a tested restore.
14	C	A risk is possible future harm; the others already happened (incidents).
15	B	2 × 3 = 6; bands are 1–2 Low, 3–4 Medium, 6–9 High.
16	D	5 cannot be made from 1–3 × 1–3.
17	A	Restore lowers how much is lost (impact); a patch lowers likelihood.
18	C	Origin-bound keys check the website automatically, so stolen details were not enough.
19	D	Know + have. A, B, C are all "know + know".
20	A	Social engineering hacks the human factor, not code.
21	B	Pretext = invented story; requested action = what causes harm.
22	C	"From the court" = authority; "ten minutes" = urgency.
23	D	Following an authorized person into a restricted area.
24	A	Text channel (smishing) + targeted (spear) + claimed identity (impersonation).
25	B	The request stayed the same; the person's state changed.
26	C	Seeing a familiar face is not passing the approval process.
27	D	Read the hostname from the right: `…campus.example`. Option A belongs to `verify-now.test`.
28	A	HTTPS also protects a connection to a deceptive site.
29	B	Money sent → 112 + bank immediately for payment suspension.
30	C	118 KISA = technical advice on suspicious SMS / malicious apps.
31	D	Identification = the account claim.
32	A	The stolen password was valid proof, so authentication passed.
33	B	Stuffing does not guess; it reuses a leaked pair.
34	C	Offline guesses never reach the login service.
35	D	A hash is one-way; the guess produced a match, nothing was decrypted.

Q	Ans	Why
36	A	A salt makes every hash different; it is stored with the hash and is not secret.
37	C	Minimum 15 characters for password-only; long, random and unique.
38	B	Another user's resource at the same level = horizontal.
39	A	Only the access the assignment needs.
40	D	A verification screen alone does not enforce the rule; the route must check MFA state.
41	B	Malware is defined by purpose and actions, not names or symptoms.
42	C	A worm spreads by itself through reachable, unpatched systems.
43	A	Disguised as something useful = Trojan.
44	D	Collects information; a keylogger captures keystrokes.
45	B	Backups must be protected, separated and tested.
46	C	Fix the cause too, or the weakness comes back.
47	A	Admin rights let malware change more and disable controls.
48	D	Contact → execution → privilege → persistence → impact → recovery.
49	B	Ransomware + worm via SMB on reachable, unpatched Windows systems.
50	D	Switches connect local devices; routers connect networks.
51	A	Only content in transit between TLS endpoints.
52	C	Segmentation limits unnecessary paths and an incident's spread.
53	B	22 = SSH, 443 = HTTPS.
54	D	DNS: `lms.example.com` → `198.51.100.20`.
55	A	An SSID only proves the advertised name.
56	C	An IP address identifies an interface; it does not prove identity.
57	B	NAT: many private addresses share one public address.
58	D	Laptop → AP → switch → printer stays local.
59	A	Name the location first; stop at certificate warnings.
60	C	Without a valid certificate for that name, TLS shows a warning.

Scoring guide: 54–60 excellent · 45–53 good, review the weeks you missed · below 45 re-read the study guide, then take the EXTRA practice bank.